6406531895289. ✖ Adds random jitter to feature values

Speech Tech

Section Id :	64065338437
Section Number :	10
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	40
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065381192
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 236 Question Id : 640653567121 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531895290. ✔ YES

6406531895291. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065381193
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 237 Question Id : 640653567122 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

1.1 MCQ *(Note: Please explain your choice)* 8 points

1. (1 point) For a given set of data points X = $\{x_1, x_2, ..., x_N\}$, log-likelihood for a GMM with k mixtures can be defined as:

   A. $\sum_{j=1}^{N} \log(\sum_{i=1}^{k} \pi_i N(x_j, \mu_i, \sigma_i^2))$

   B. $\log(\prod_{j=1}^{N} \sum_{i=1}^{k} \pi_i N(x_j, \mu_i, \sigma_i^2))$

   C. both A and B

   D. None of these

2. (1 point) Given a HMM with N hidden states and M observation symbols, let $b_j(k)$ represent the probability of observing event j when in state k. Also, let π_j represent the probability of being in state j initially (π_j's represent the initial state probabilities). Which of the following statements is True?

   A. $\sum_{k=1}^{M} b_j(k) \geq 1$ and $\sum_{j=1}^{N} \pi_j \leq 1, \forall 1 \leq j \leq N.$

   B. $\sum_{j=1}^{N} b_j(k) = 1$ and $\sum_{j=1}^{N} \pi_j = 1, \forall 1 \leq k \leq M.$

   C. $\sum_{k=1}^{M} b_j(k) = 1$ and $\sum_{j=1}^{N} \pi_j = 1, \forall 1 \leq j \leq N.$

   D. $\sum_{j=1}^{N} b_j(k) \geq 1$ and $\sum_{j=1}^{N} \pi_j \leq 1, \forall 1 \leq k \leq M.$

3. (1 point) If $f(z) = tanh(z)$, then the derivative of $f(z)$ will be?

   A. $f(z)$

   B. $1 - f(z)$

   C. $1 - f^2(z)$

   D. $f^2(z)$

4. (1 point) Recurrent Neural Networks(RNN) with basic RNN cells exhibit which of the following problems?

   A. Vanishing gradient problem

   B. Exploding gradient problem

   C. both (A) and (B)

   D. None of the above

Options :

6406531895292. ✔ I have written answers on the answer sheets

6406531895293. ✖ Not applicable

Question Number : 238 Question Id : 640653567123 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

5. (1 point) Which of the following ideas is used by LSTM-based RNN networks to capture long range dependencies?

   A. Pooling
   B. Convolution
   C. Gates
   D. None of the above

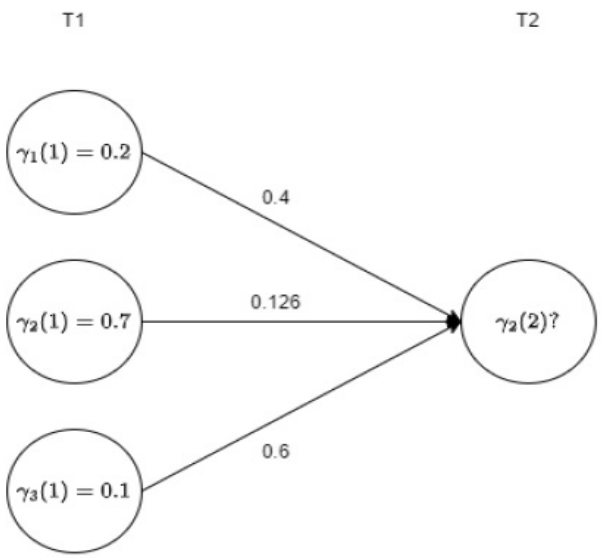
6. (1 point) State whether the following statement is true or false: A Transformer Encoder layer can compute the hidden representation of all time steps simultaneously.

   A. True
   B. False

7. (1 point) Which of the following is correct according to Bayes rule?

   A. $P(A|B) = \frac{P(A)}{P(B)}$
   B. $P(A|B) = \frac{P(A \cap B)}{P(A)}$
   C. $P(A|B) = \frac{P(A \cap B)}{P(B)}$
   D. $P(A|B) = \frac{P(A \cup B)}{P(B)}$

8. (1 point) Consider the Fig, and determine $\gamma_2(2)$



   A. 0.08
   B. 0.0882
   C. 0.006
   D. 0.2282

Options :

6406531895294. ✔ I have written answers on the answer sheets

6406531895295. ✖ Not applicable

Sub-Section Number :	3
Sub-Section Id :	64065381194
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 239 Question Id : 640653567124 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 14

Question Label : Multiple Choice Question

1.2 Subjective Questions 32 points

1. (2 points) CNN is a popular building block of many state-of-the-art neural networks; Discuss CNNs in detail.

2. (3 points) Which of the following are valid activation functions you could use to train neural networks. Draw graphs representing the result of using activation function at the input, similar to graphs for activation functions shown in the class. Justify your answer.

   - $f(x) = max(0, x)$
   - $f(x) = 0.2x + 1$
   - $f(x) = \begin{cases} x & if \ x > 0.5 \\ -1 & else \end{cases}$

3. (3 points) Explain the difference between local and global minima. Discuss the steps suggested in literature to avoid problem of getting stuck at a local minima.

4. (3 points) In class we discussed a class of models coined as 'Language Models', these are useful for capturing the underlying structure of the language. Explain Language Modelling and N-gram model in detail.

5. (3 points) We saw that for a given decoder timestep t, the attention weights over the S encoder states (timesteps) are constrained to sum up to 1 ($i.e., \sum_{t=1}^{S} \alpha_{it} = 1$). However, there is no constraint on how the attention weights behave across timesteps. This could result in a situation where the attention weights are always concentrated on the same set of encoder states across different decoder timesteps. This may be undesirable (for example, in the case of translation this might mean that some words in the source sentence never get attention). Suggest a way to tweak the loss function to explicitly prevent this problem?

Options :

6406531895296. ✔ I have written answers on the answer sheets

6406531895297. ✖ Not applicable

Sub-Section Number :	4
Sub-Section Id :	64065381195
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 240 Question Id : 640653567125 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 18

Question Label : Multiple Choice Question

6. (3 points) To decode the utterance of a speaker, alignment is an important step. Discuss alignment in detail and how CTC helps achieve this.

7. (5 points) We saw that classical encoder-decoder models face the problem of 'Fixed Encoder Representation'; Discuss how the attention mechanism is used to solve this issue. Also discuss briefly, how the attention mechanism of the LSTM-based encoder-decoder model differs from the attention mechanism introduced in the Transformer.

8. (5 points) Consider a weather prediction based HMM problem with three hidden states: *Sunny, Cloudy and Rainy*. There are four different observable states that depend on the hidden states, namely: *Dry, Dryish, Damp, and Soggy*. The transition and observation probabilities are shown in fig. Given initial Rainy weather condition on Day0. It was observed that the weather was *Soggy* followed by *Dry* for the next two days. Compute the best possible weather forecast sequence which best explains these two observations (*hint: Use Viterbi Algorithm*).

Weather \ Grass	Dry	Dryish	Damp	Soggy
Sunny	0.60	0.20	0.15	0.05
Cloudy	0.25	0.25	0.25	0.25
Rainy	0.05	0.10	0.35	0.50

Yesterday \ Today	Sunny	Cloudy	Rainy
Sunny	0.5	0.375	0.125
Cloudy	0.25	0.125	0.625
Rainy	0.25	0.075	0.675

9. (5 points) We discussed Bidirectional Encoder Representations from Transformer (BERT) as a way to obtain embeddings, which can be helpful in many NLP tasks. List the mechanism used for training the BERT model and highlight differences with the *Word2Vec* training.

Options :

6406531895298. ✔ I have written answers on the answer sheets

6406531895299. ✖ Not applicable

Market Research

Section Id :	64065338438
Section Number :	11
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	4
Number of Questions to be attempted :	4
Section Marks :	50
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065381196
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 241 Question Id : 640653567126 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : MARKET RESEARCH (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.